CHEMISTRY ADAPTLY

# ENERGY CHANGES – EXOTHERMIC & ENDOTHERMIC

## KEY VOCABULARY

**Exothermic:** transfers energy to the surroundings (gets hotter).
**Endothermic:** takes in energy from the surroundings (gets colder).
**Surroundings:** everything around the reaction (e.g. the water).
**Combustion:** burning — an exothermic reaction.

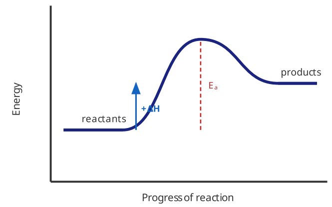
## COMMON MISCONCEPTIONS

~~Exothermic reactions feel cold.~~
✓ Exothermic releases heat — feels hot.

~~Endothermic reactions destroy energy.~~
✓ Energy is taken in, not destroyed.

~~All reactions release energy.~~
✓ Some (endothermic) absorb energy.

## ENDOTHERMIC PROFILE




Products are **higher** than reactants — energy is **taken in**.

## 1 DEFINE EXOTHERMIC

State what is meant by an **exothermic reaction**. **[1 mark]**

## 2 CLASSIFY

Classify combustion and thermal decomposition as exo- or endothermic. **[2 marks]**

## 3 TEMPERATURE CHANGE

**Explain** why the temperature of the surroundings falls in an endothermic reaction. **[2 marks]**

## 4 EVERYDAY USES

**Give** a use of an exothermic and an endothermic change. **[2 marks]**

## 5 FILL THE GAPS

Complete the sentence using the word bank. **[2 marks]**

*An exothermic reaction _____ energy, so products are at a _____ energy level.*

releases | takes in | higher | lower

## 6 INTERPRET A PROFILE

A reaction profile shows products below reactants. **State** the type of reaction and why. **[2 marks]**

***TIP:*** *Exo = exit heat (hotter). Endo = energy in (colder).*

**✓ Think: Why does an exothermic reaction have products at a lower energy than reactants?**